



MD-100 Windows Client

Module 04: Configuring networking on Windows clients

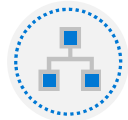
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Module agenda



Configure IP network connectivity



Explore name resolution



Configure wireless network connectivity



Explore remote access



Employ remote management

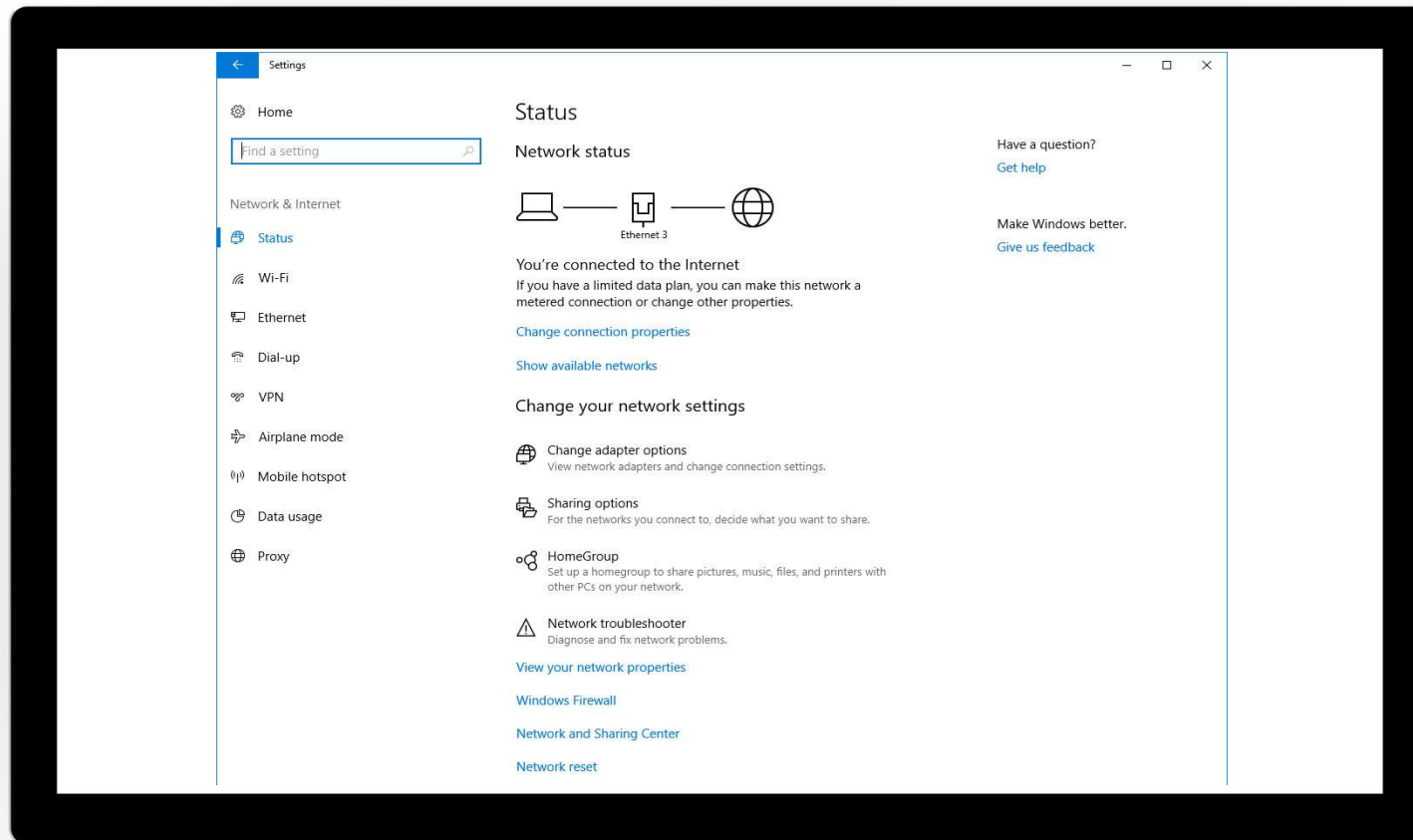
Lesson 01: Configure IP network connectivity



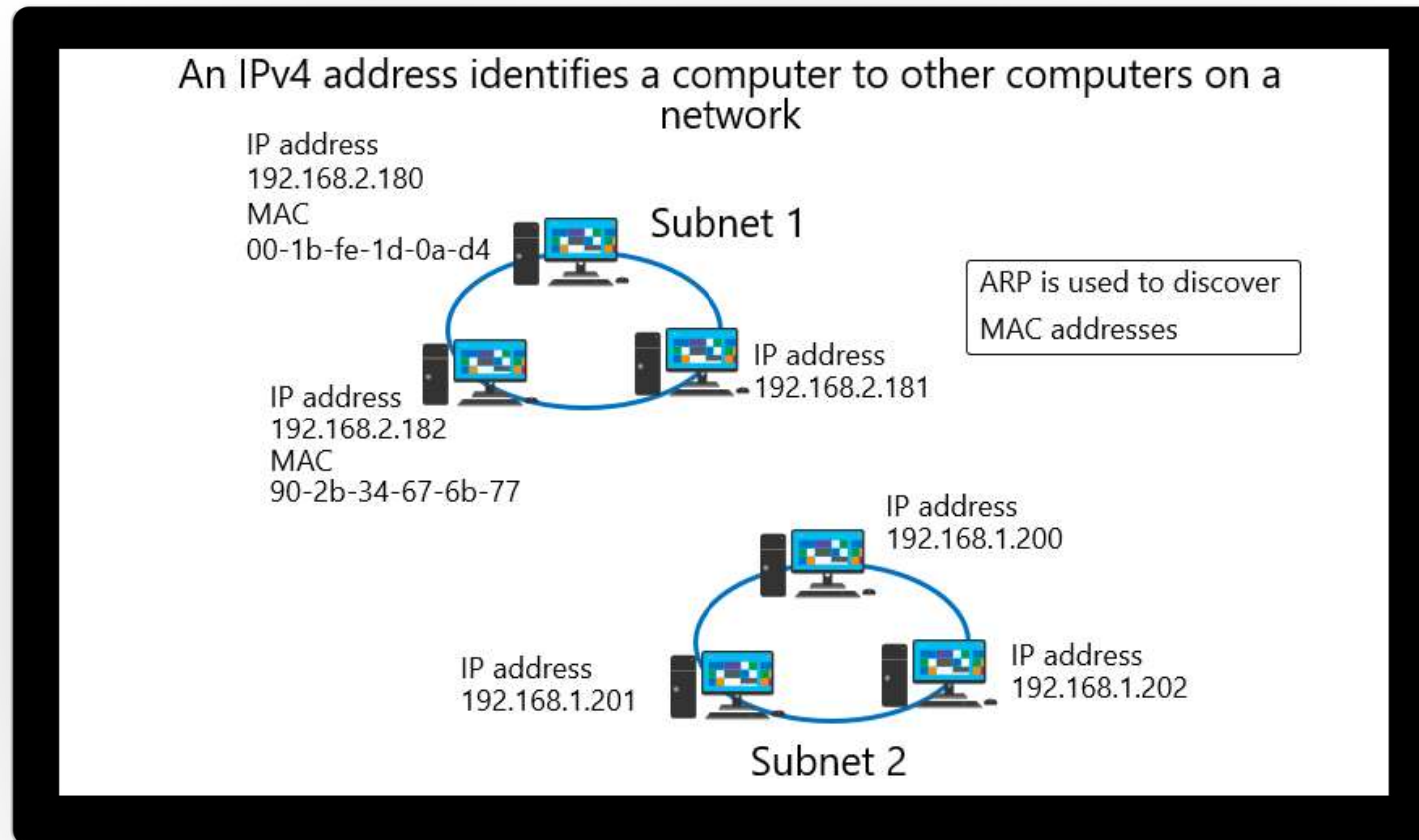
Lesson 1: Configure IP network connectivity

- Explore Windows network configuration
- Explore IPv4 settings
- Define subnets for IPv4 networks
- Explain public and private IP addressing
- Implement automatic IPv4 addressing
- Examine tools for configuring network settings
- Examine tools for troubleshooting network connections
- Examine IPv6 addressing
- Explore IPv6 in Windows client

Explore Windows network configuration

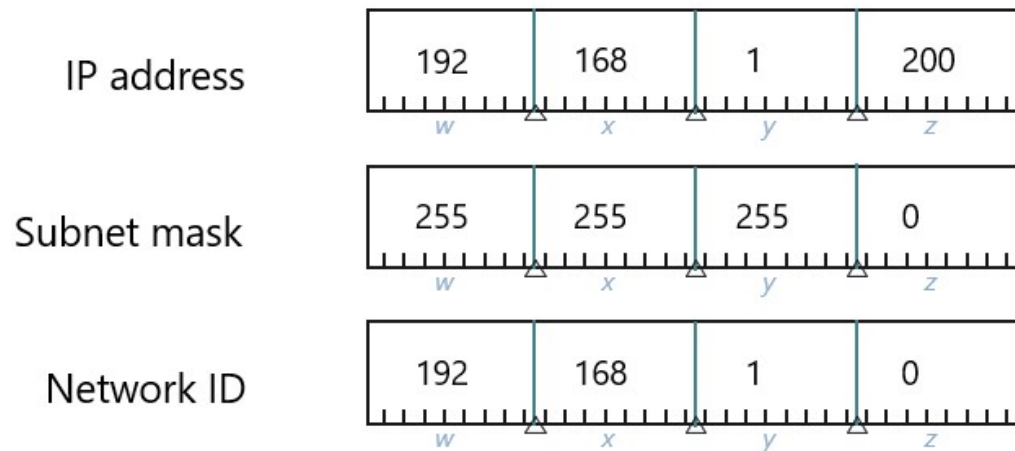


Explore IPv4 settings



Define subnets for IPv4 networks

A subnet mask specifies which part of an IPv4 address is the network ID and which is the host ID



Define subnets for IPv4 networks

A default gateway is a device, usually a router, on a TCP/IP internetwork that forwards IP packets to other subnets



Subnet 1

Default gateway



Subnet 2



Router



Explain public and private IP addressing

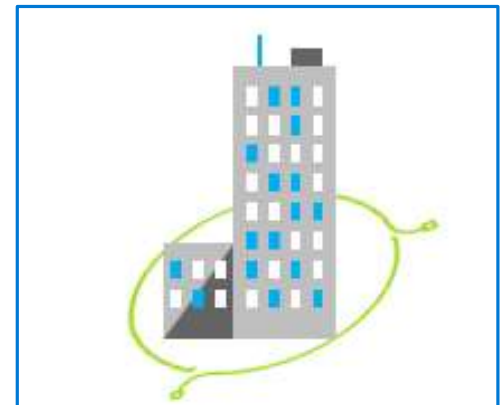
Public

- Required by devices and hosts that connect directly to the internet
- Must be unique
- Routable on the internet
- Must be assigned by the IANA

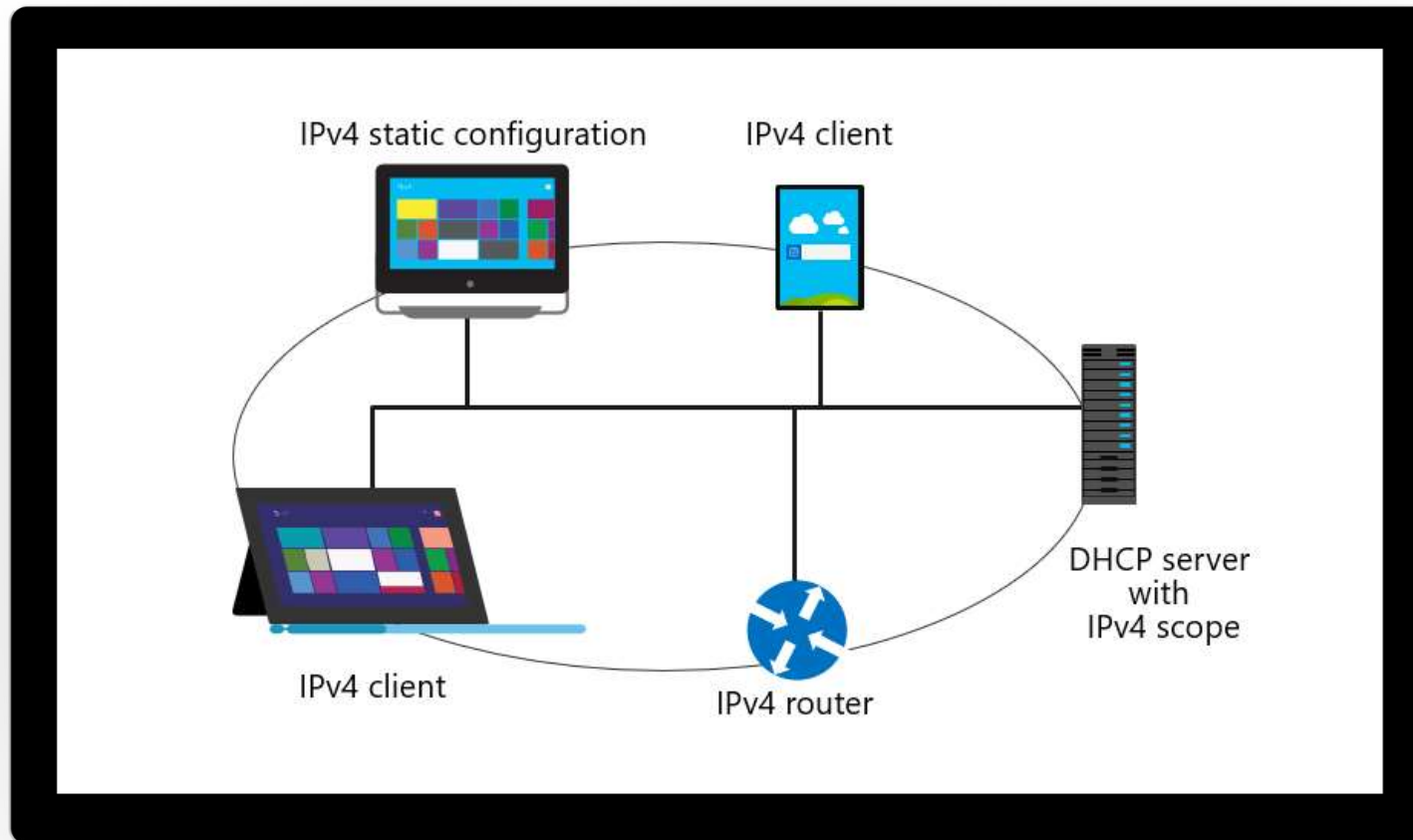


Private

- Non-routable on the internet
- Assigned locally by organization
- Must be translated to access the internet



Implement automatic IPv4 addressing



Examine tools for configuring network settings

Windows provides a number of tools to enable, configure, and manage network connections and settings:

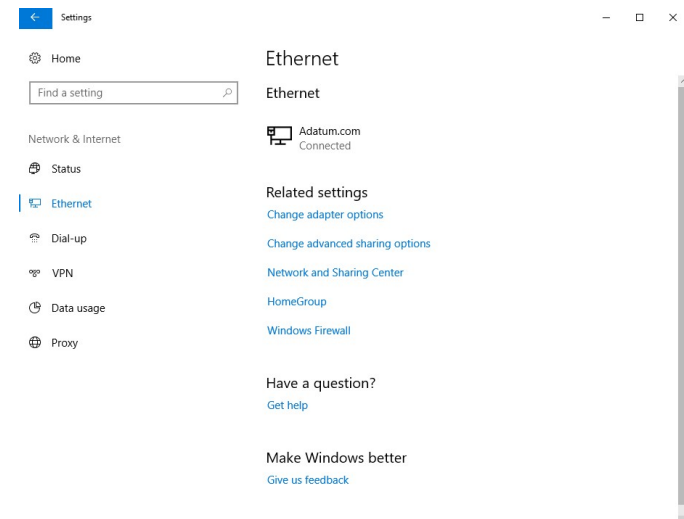
- Network & Internet
- Network and Sharing Center
- Network Setup Wizard
- Windows PowerShell
- Netsh

```
Administrator: Windows PowerShell

PS C:\Users\Administrator> get-netipaddress

IPAddress      : fe80::5efe:172.16.0.40%2
InterfaceIndex  : 2
InterfaceAlias  : isatap.{5960286C-A9CF-426F-B830-50890B351C57}
AddressFamily   : IPv6
Type            : Unicast
PrefixLength    : 128
PrefixOrigin    : WellKnown
SuffixOrigin    : Link
AddressState    : Deprecated
ValidLifetime   : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource    : False
PolicyStore     : ActiveStore

IPAddress      : fe80::546c:6258:f4bd:cdf6%3
InterfaceIndex  : 3
InterfaceAlias  : Ethernet
AddressFamily   : IPv6
Type            : Unicast
PrefixLength    : 64
PrefixOrigin    : WellKnown
SuffixOrigin    : Link
AddressState    : Preferred
ValidLifetime   : Infinite ([TimeSpan]::MaxValue)
PreferredLifetime : Infinite ([TimeSpan]::MaxValue)
SkipAsSource    : False
```



Examine tools for troubleshooting network connections

Tool	Purpose
Event Viewer	View errors relating to network activity
Windows Network Diagnostics	Diagnose and resolve network problems
IPConfig	Display IP configuration information and control the DNS resolver cache
Ping and Pathping	Verify basic IP connectivity
Tracert	Verify a routing path
NSLookup	Test name resolution
Windows PowerShell	Configure and troubleshoot network-related settings

Examine IPv6 addressing

IPv6 address types:

- Unicast is used for one-to-one communication between hosts
- Multicast is used for one-to-many communication between computers that are defined as using the same multicast address
- Anycast is used for locating services or the nearest router

IPv6 unicast address types:

- Global unicast is globally routable and reachable on the IPv6 portion of the internet
- Link-local is used when communicating with neighboring hosts on the same link
- Unique local unicast is equivalent to IPv4 private address spaces

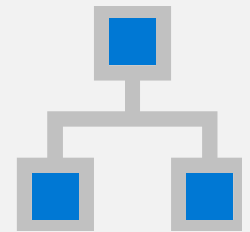
Explore IPv6 in Windows client

Benefits of IPv6 include:

- Large address space
- Hierarchical addressing and routing infrastructure
- Stateless and stateful address configuration
- Required support for IPsec
- Restored end-to-end communication
- Prioritized delivery
- New protocol for neighboring node interaction
- Extensibility

Knowledge Check

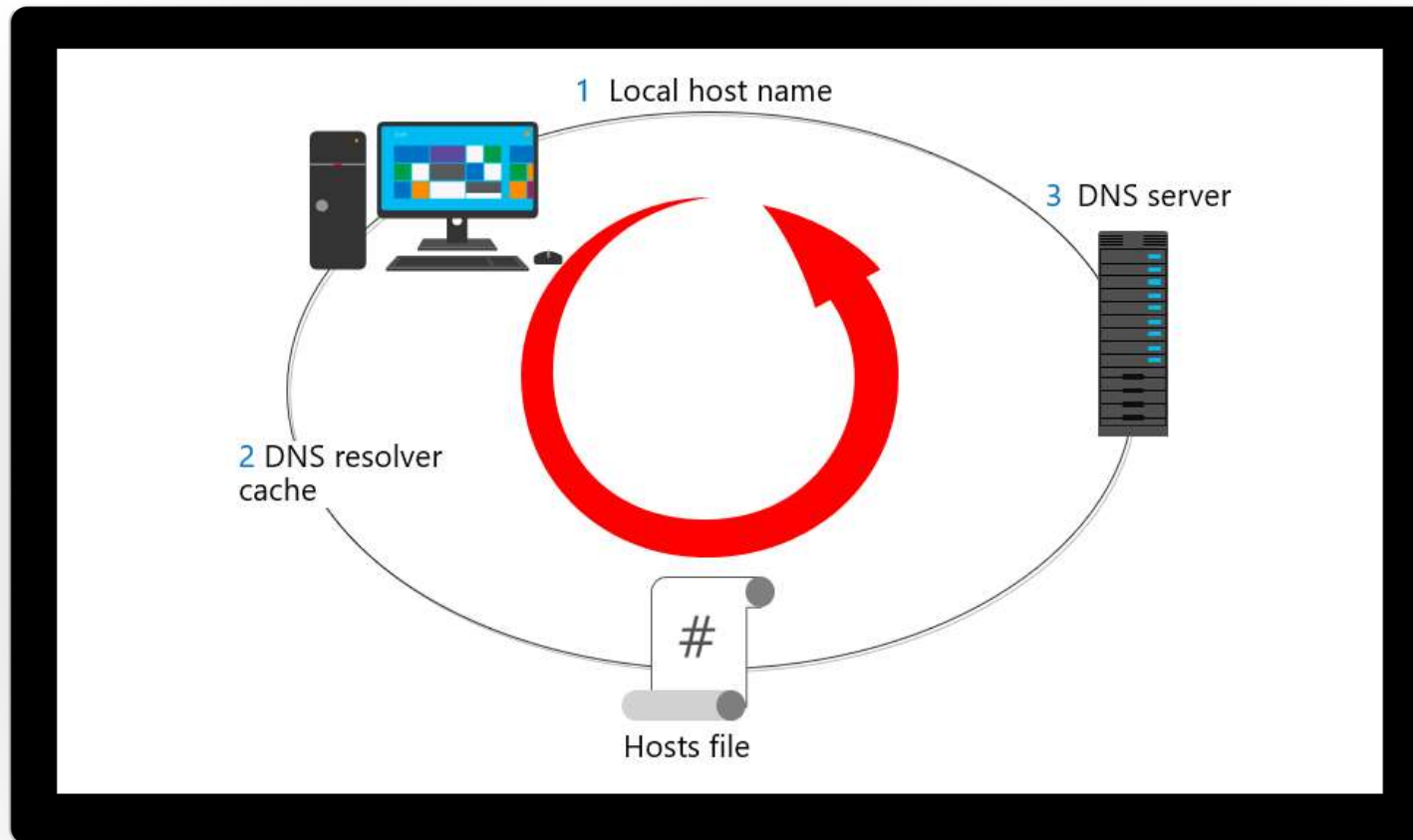
Lesson 02: Explore name resolution



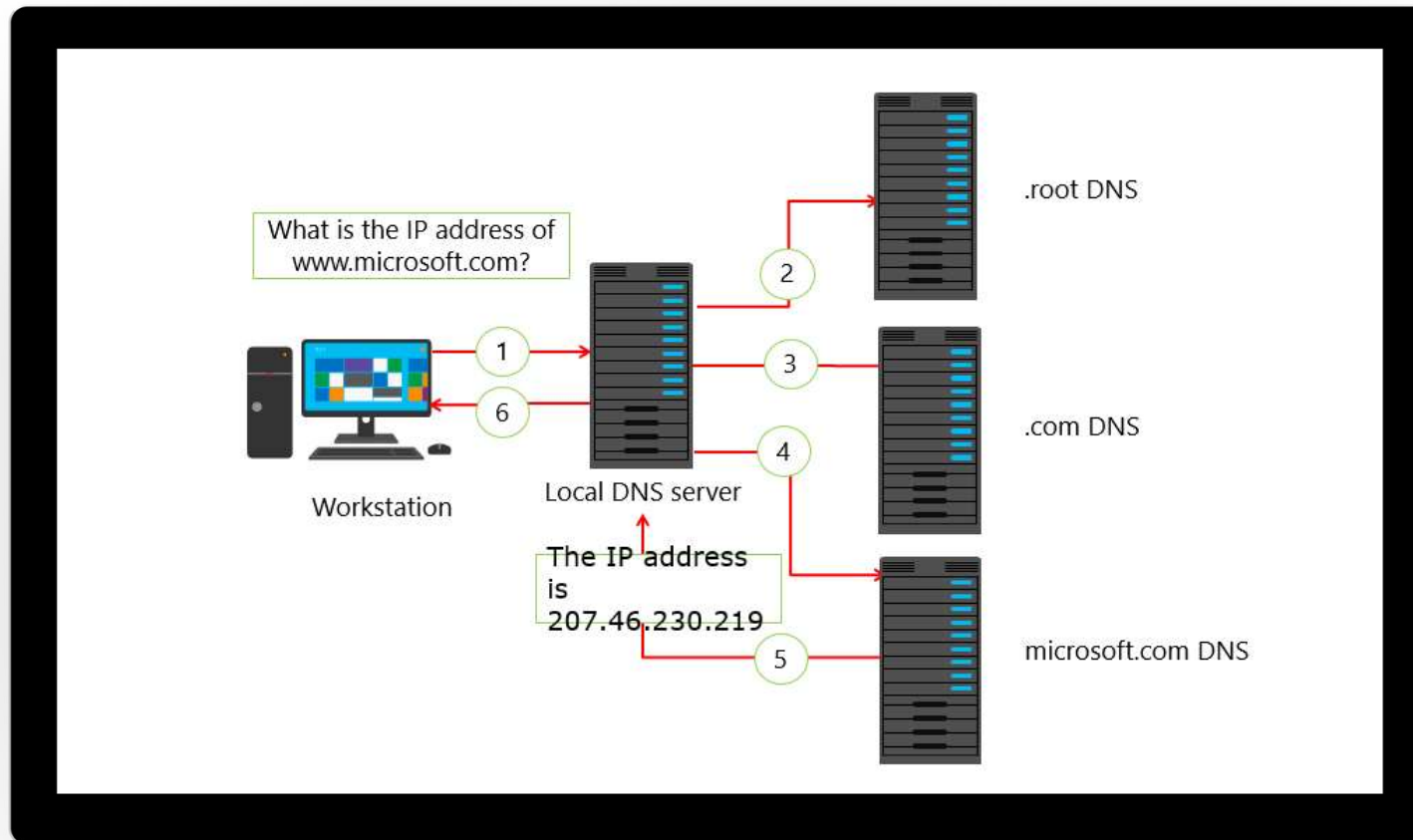
Lesson 2: Explore name resolution

- Explain how name resolution works
- Explore the domain name system
- Troubleshoot name resolution

Explain how name resolution works



Explore the domain name system



Troubleshoot name resolution

To troubleshoot name resolution:

- Clear DNS resolver cache
- Verify connectivity to the remote host by IP address
- Verify connectivity to the remote host by hostname
- If the test is unsuccessful, edit the Hosts text file
- Repeat step 3. Name resolution should now be successful
- Remove the entry that you added to the Hosts file, and clear the resolver cache
- Verify name resolution function. At the command prompt, type one of the following commands:

```
Nslookup.exe -d2 LON-cl1.adatum.com. > filename.txt
```

```
Resolve-dnsname lon-cl1.adatum.com. > filename.txt
```

Knowledge Check

Lesson 03: Configure wireless network connectivity

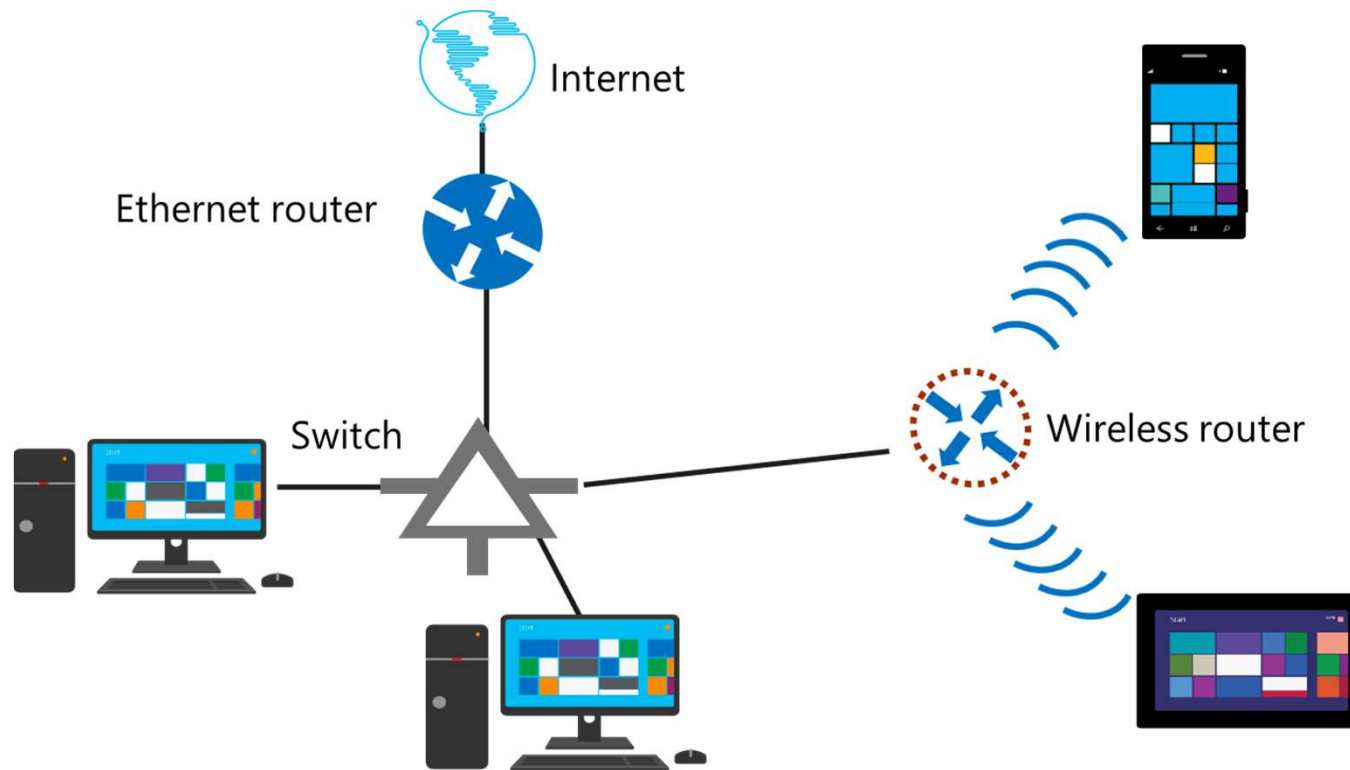


Lesson 3: Configure wireless network connectivity

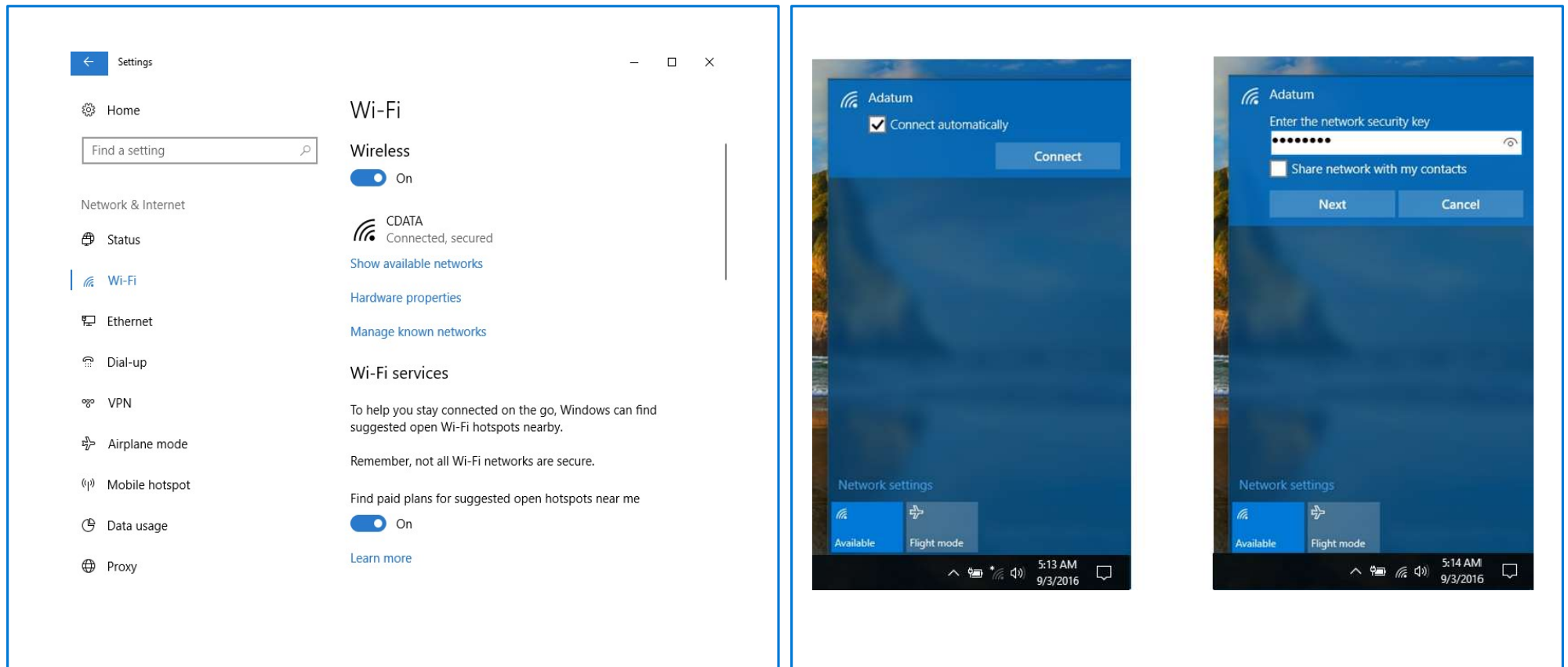
- Examine wireless network technologies
- Configure wi-fi settings and profiles
- Configure wireless display with Miracast

Examine wireless network technologies

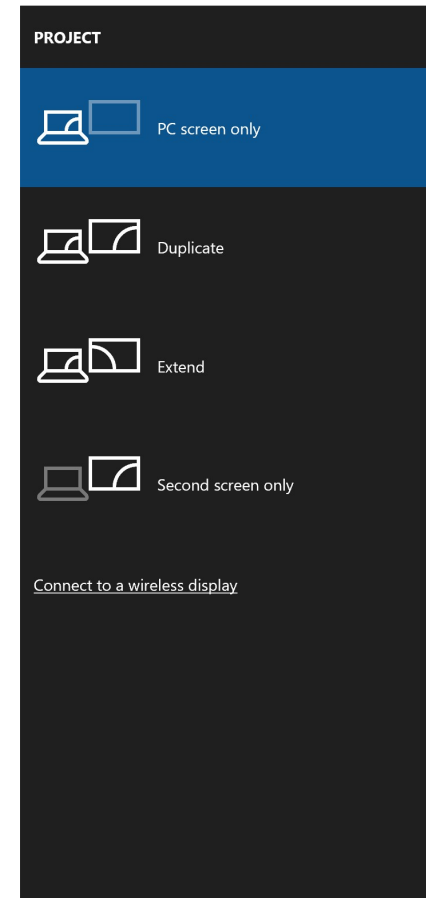
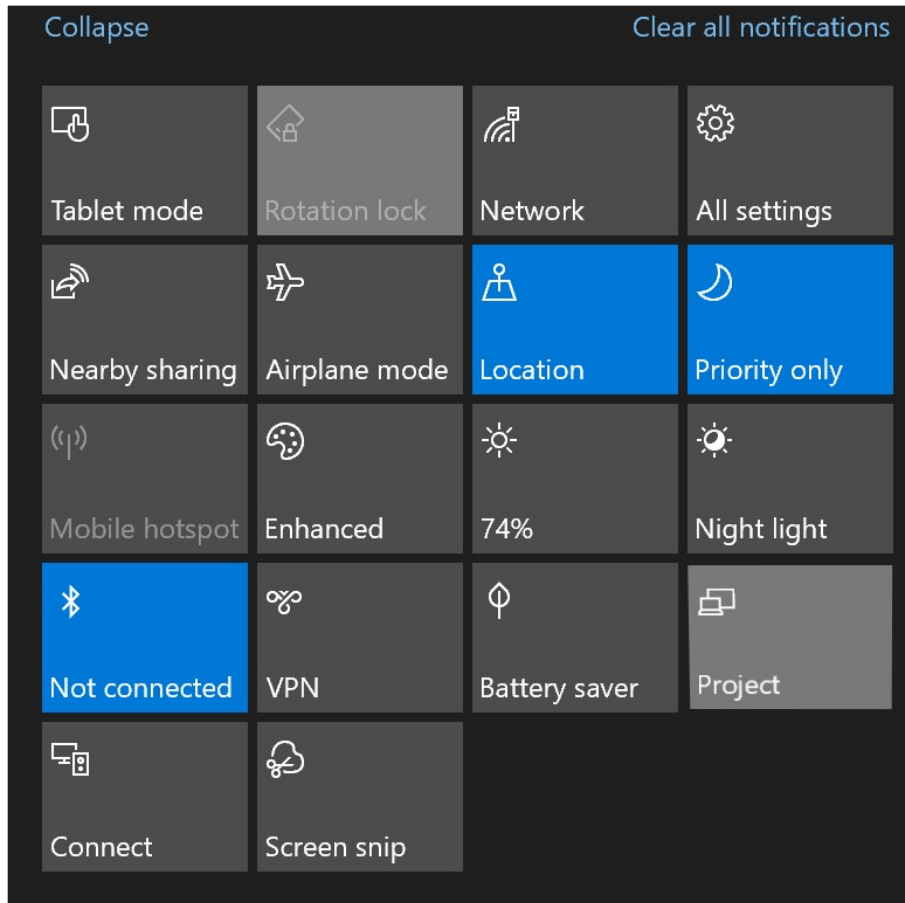
Modern networks are often a combination of wired and wireless networks



Configure wi-fi settings and profiles

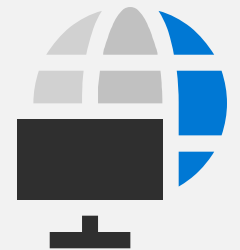


Configure wireless display with Miracast



Knowledge Check

Lesson 04: Explore remote access

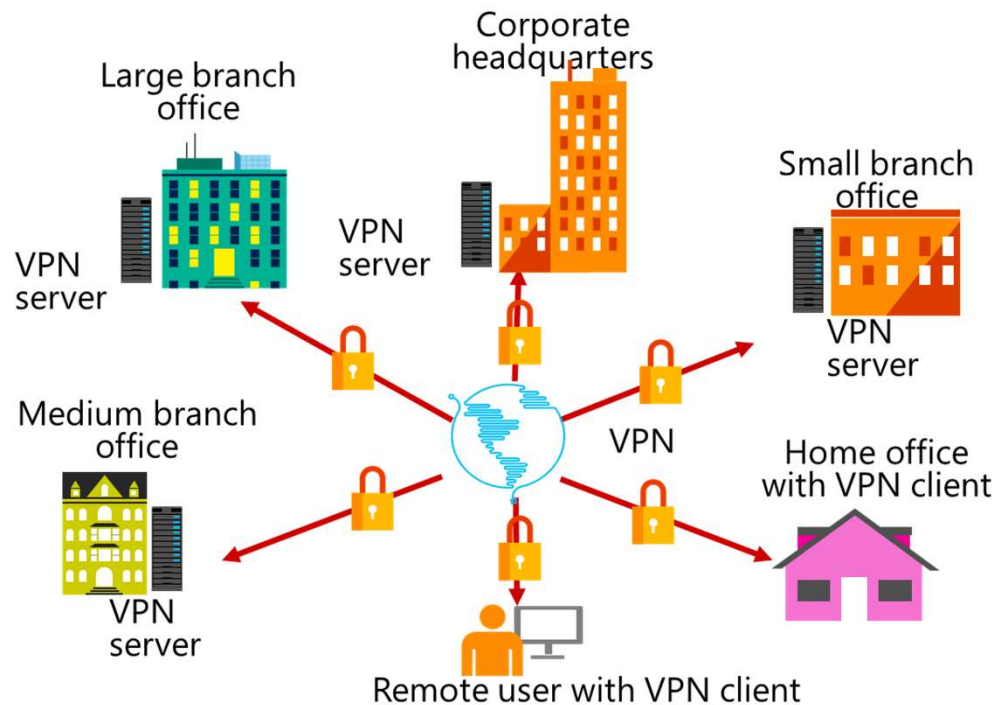


Lesson 4: Explore remote access

- Explore virtual private networks
- Create a VPN connection in Windows client
- Describe Always On VPN

Explore virtual private networks

A VPN provides a point-to-point connection between components of a private network, through a public network such as the internet



Create a VPN connection in Windows client

Add a VPN connection

VPN provider
Windows (built-in) ▼

Connection name
Corporate HQ

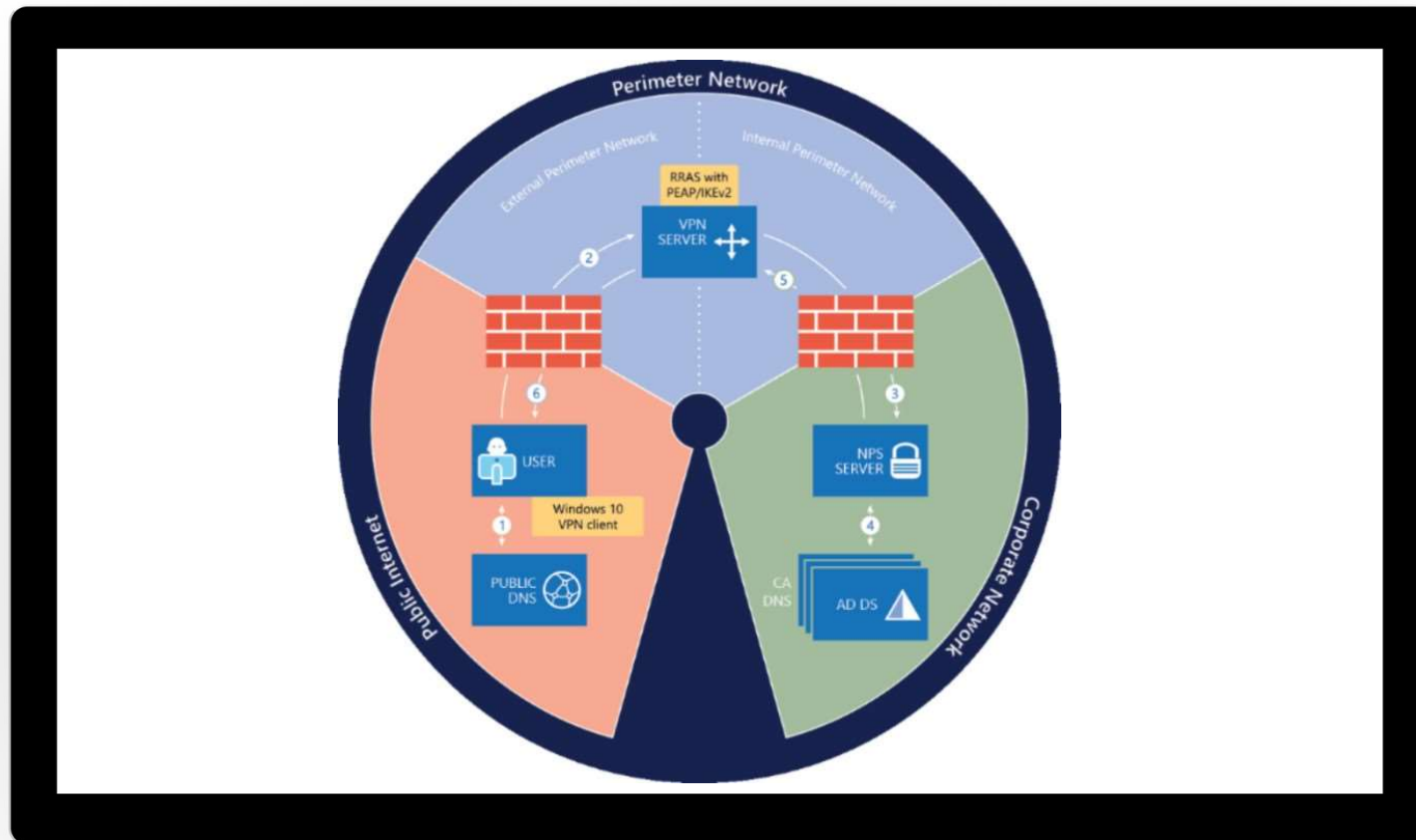
Server name or address
lon-dc1.adatum.com

VPN type
L2TP/IPsec with certificate ▼

Type of sign-in info
User name and password ▼

Save Cancel

Describe Always On VPN



Knowledge Check

Lesson 05: Employ remote management

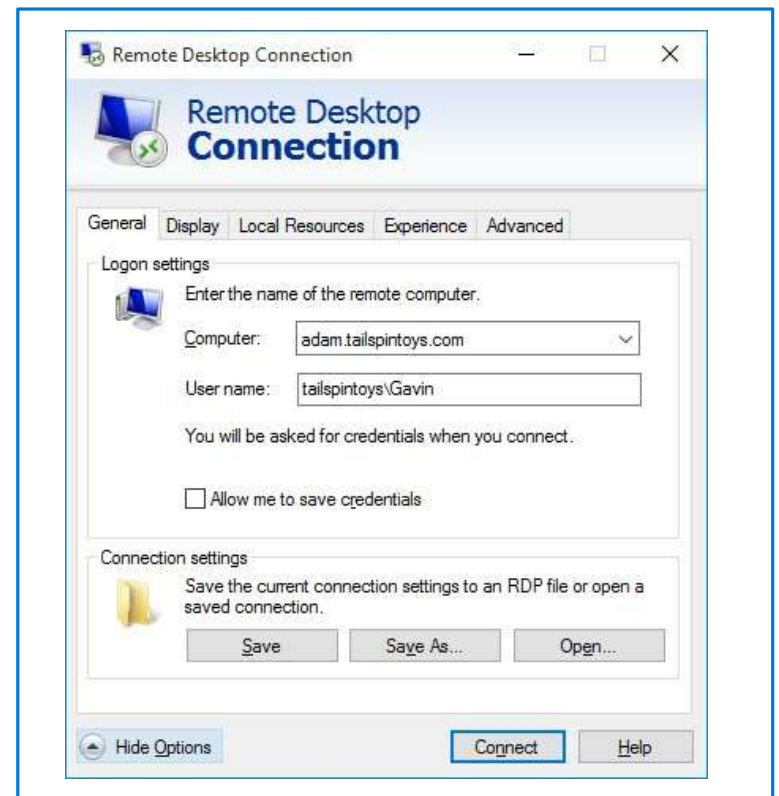
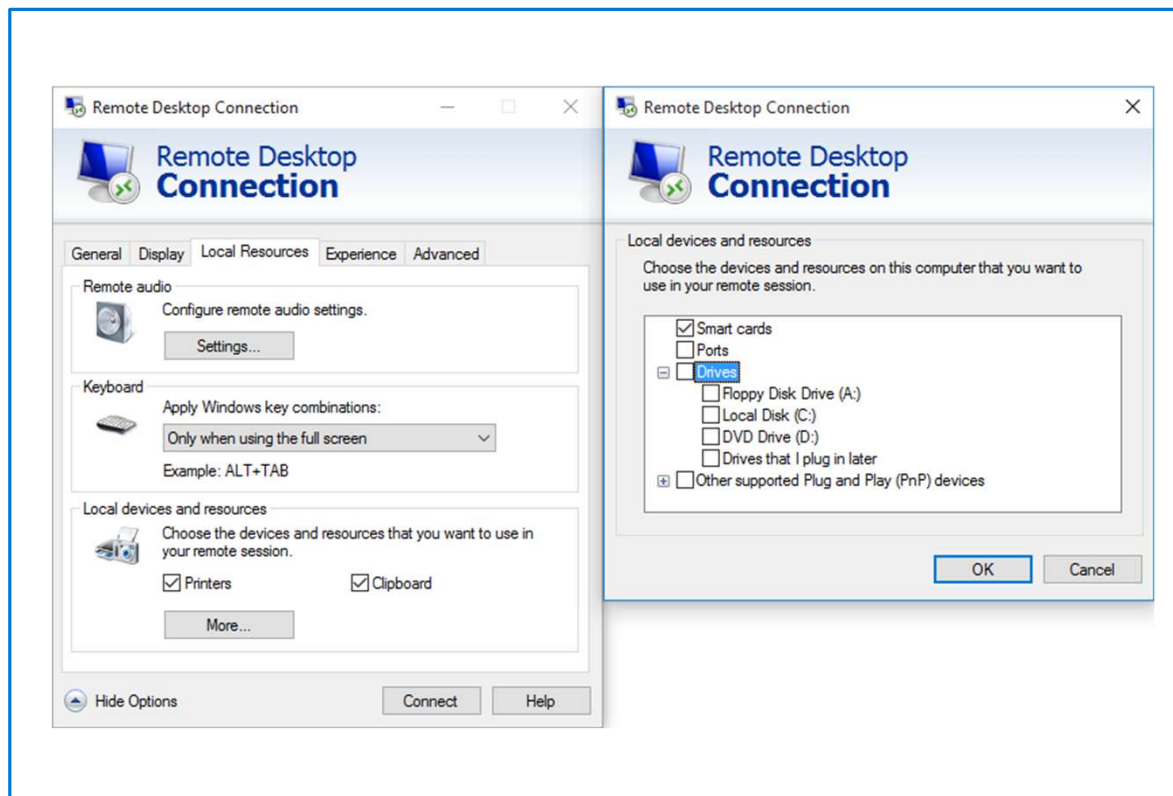


Lesson 5: Employ remote management

- Examine Remote Desktop
- Use Remote Assistance
- Examine Remote Desktop services
- Use remote commands in Windows PowerShell
- Describe Windows Admin Center

Examine Remote Desktop

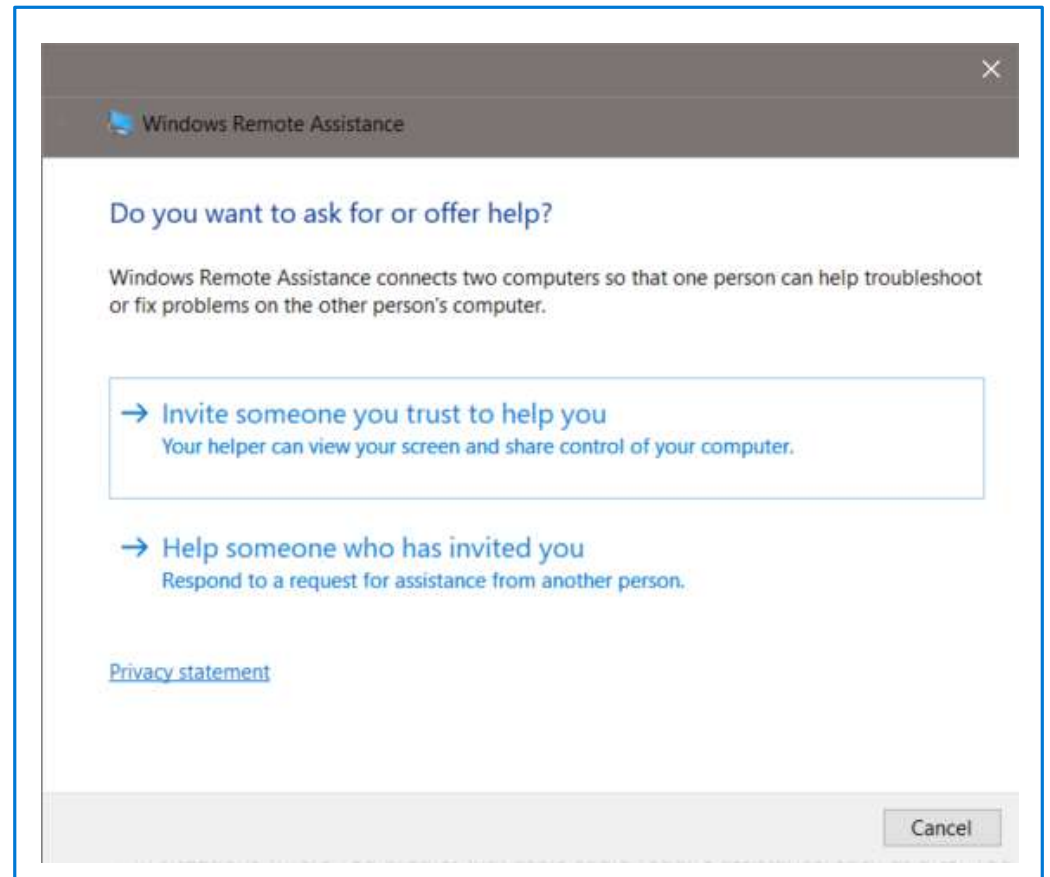
- A built-in feature that people use to access computers from other devices that support RDP
- Used by administrators to connect to remote devices for management purposes



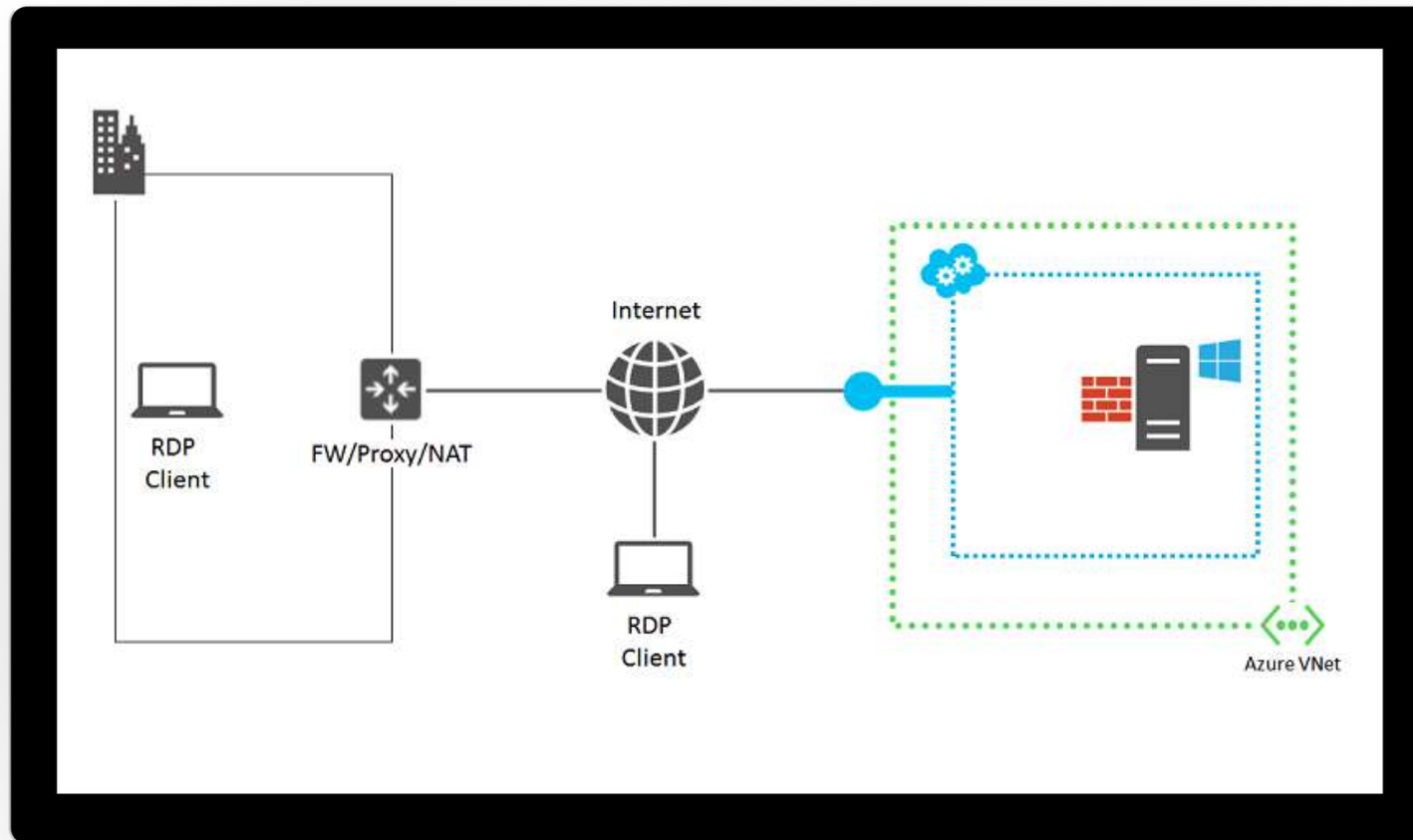
Use Remote Assistance

Remote Assistance enables you to:

- View a remote desktop
- Create a chat session
- Control desktops remotely



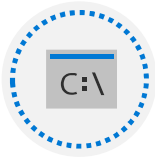
Examine Remote Desktop services



Use remote commands in Windows PowerShell



Remoting features of Windows PowerShell are built on Windows Remote Management



Run an individual command or create a persistent connection or session to run a series of commands



To enable remoting, use the following procedure:

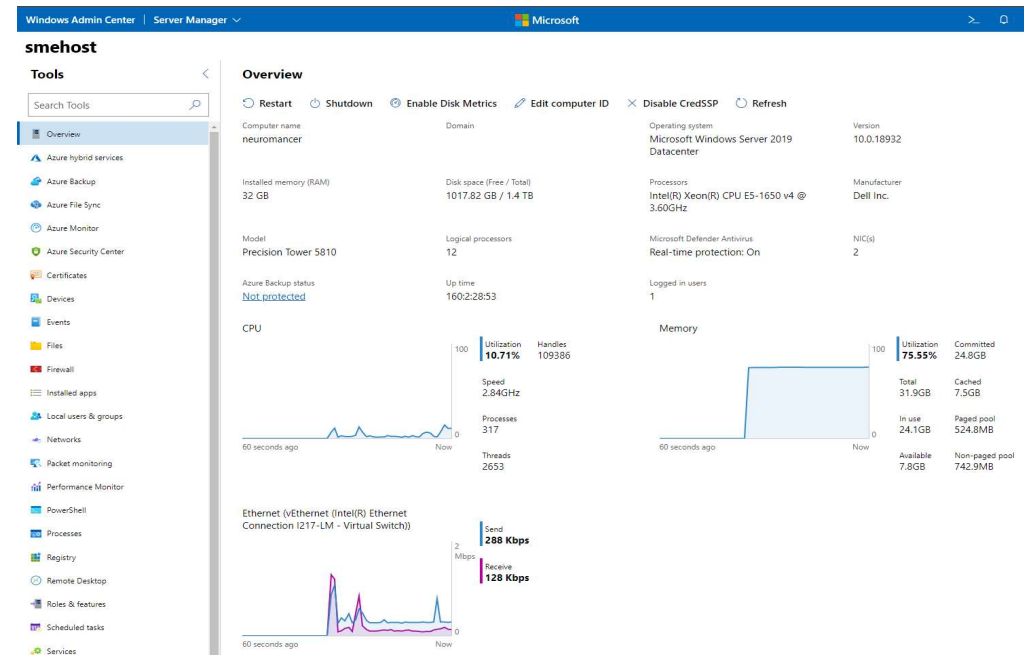
Verify the status of the Windows Remote Management service: `Winrm quickconfig`

Enable remoting: `Enable-PSremoting -force`

Describe Windows Admin Center

Remote manage servers and clients

- Display resource and utilization
- Device management
- Event viewer
- Local users and groups
- View/end processes
- Scheduled tasks
- Windows Services



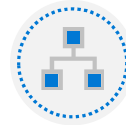
Knowledge Check

Practice Labs

Configuring Network Connectivity



Configuring and Testing Name Resolution



Administering Windows Client Using Remote Management



Module Review

